Venkata Lakshmi Parimala Pasupuleti

Backend Engineer · Production GenAI Systems · RAG Architectures · LLM Evaluation · Agentic Tool-Calling · Python · FastAPI · AWS · 9yr Production

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PROFESSIONAL SUMMARY

Backend engineer with 9+ years of production systems experience and a specialization in GenAI systems built to be trusted - not prototypes or thin wrappers, but retrieval architectures, evaluation infrastructure, and agentic pipelines that hold up under real operating conditions. AgentProbe is the direct proof: I designed and built a production AI evaluation platform from scratch - 5-metric quality engine measuring correctness, groundedness, hallucination rate, latency, and token efficiency; LLM-as-Judge scoring with pass@k multi-trial harness for variance detection in non-deterministic outputs; structured tracing across all agent steps; and a CI-integrated regression gate blocking quality degradation before deployment. This is exactly the evaluation and testing infrastructure that catches failure modes and prevents regressions in AI behavior - built because I understood what "AI that can be trusted" requires at the systems level. LeaseI demonstrates the agentic pattern under high-consequence constraints: an 8-agent LangGraph system with deterministic HITL guardrails where the LLM reasons and extracts but deterministic logic makes every final regulated call - the architecture you need when AI mistakes carry real consequences. At Entrata I owned mission-critical Python backend systems for 7 years at 99.5% uptime - complex compliance logic, async distributed processing with RabbitMQ, and zero compliance violations across full ownership. I'm based in Arlington, VA. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Production GenAI & Agent Systems: Python (9yr), LangGraph, LangChain, RAG Architectures, Vector Databases (ChromaDB), Embedding Pipelines, Semantic Search, Retrieval Optimization, Agentic Systems, Tool Calling, Multi-Agent Orchestration, LLM Integration, Multi-Provider LLM Routing, Prompt Engineering, LLM Evaluation, LLM-as-Judge, Hallucination Detection, Groundedness Scoring, pass@k Evaluation Harness, Failure Mode Analysis, Regression Testing, Output Validation, HITL Guardrails, Confidence Scoring, Anthropic API, OpenAI API, Ollama, FastAPI

Backend, APIs & Distributed Systems: FastAPI, Flask, REST APIs, API Design, Distributed Systems, Async Processing, RabbitMQ, Microservices, PostgreSQL, MySQL, Oracle, Redis, SQL, NoSQL, High-Throughput Systems, Performance Optimization, Query Optimization, Idempotency, Reliability Engineering, Observability, Structured Logging, Distributed Tracing, Monitoring, Alerting

Cloud, Infrastructure & Engineering Excellence: AWS (S3, EC2, RDS, Glue, Lambda), Docker, CI/CD (GitHub Actions), Automated Testing, Unit Testing, Performance Testing, Incident Response, Root-Cause Analysis, Runbooks, Design Documents, Architecture Decisions, Technical Mentoring, Code Reviews, Agile/Scrum, Linux, Bash, Git

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Production GenAI Systems

- Designed and deployed a production GenAI system (LangGraph agentic pipeline, FastAPI backend, Python) as sole technical owner - built RAG retrieval layer, HITL guardrails for regulated decisions, evaluation pipeline with measurable quality outcomes, and CI validation gate; delivered to production in weeks with performance metrics driving iteration throughout.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977) on AWS; applied Isolation Forest anomaly detection across 3 research pipelines surfacing failure modes manual review missed; designed observability dashboards improving comprehension 40%; mentored 600+ students in Python, SQL, and distributed systems.

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Production systems ownership at scale** - owned mission-critical Python backend systems for 7 years at 99.5% uptime serving 100K+ users and 1M+ property records; complex compliance domain logic (eligibility, rent calculations, HUD regulatory imports) with zero violations across full ownership period; RabbitMQ async event-driven distributed processing with retry logic and idempotency guarantees; 200+ PostgreSQL/MySQL/Oracle query optimizations (30% performance gain); Docker microservices redesign (50% scalability improvement).
- **Observability, reliability & incident response** - built structured logging, monitoring, and alerting infrastructure for production oversight of high-stakes compliance systems; authored runbooks; led incident response with RCA across 7 years; built internal debugging tools reducing triage time 40%; kept bug count below 10 across 20+ production modules over 120+ Agile sprints; GitHub Actions CI/CD with automated regression and performance testing; AWS S3/EC2/RDS/Glue infrastructure.
- **Architecture & technical leadership** - wrote design documents and architecture decisions for 20+ production modules; led projects spanning multiple engineers; mentored 5+ engineers; 300+ peer code reviews; Python/Bash automation eliminating 80% of manual operational workload.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (Python, PostgreSQL) for production SaaS; data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: Production AI Evaluation Platform - Catching Failure Modes Before They Ship

LLM Evaluation · RAG · Agentic · Regression Gate · Observability · Open Source

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · Anthropic API · OpenAI API · Ollama | github.com/venkatalakshmiparimala/AgentProbe

The evaluation and testing infrastructure the JD describes: built to measure output quality, surface failure modes, and prevent regressions so that AI behavior can be trusted. **Evaluation engine:** 5-metric quality measurement (correctness, groundedness, hallucination rate, latency, token efficiency); LLM-as-Judge scoring calibrated against expected outputs; pass@k multi-trial harness surfacing variance in non-deterministic AI outputs - the statistical framework for detecting unreliable behavior before it reaches production. **RAG architecture:** ChromaDB vector database with embedding pipeline and semantic retrieval; retrieval quality scored independently from generation quality. **Regression gate:** CI integration (GitHub Actions) blocks deployment when quality metrics regress - zero regressions across 1,240+ test cases. **Observability:** structured distributed tracing across all agent steps with span capture; monitoring and alerting on quality drift. Multi-provider LLM routing: Anthropic, OpenAI, Ollama. Open-source, architecturally substantive - not a prototype.

Leasel: Agentic System with HITL Guardrails - AI Trusted in High-Consequence Decisions

Agentic · Tool-Calling · RAG · HITL · High-Consequence · Production

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · SQLite · MCP · Anthropic API | github.com/venkatalakshmiparimala/Lease_complaince_rag

Agentic AI system built for high-consequence regulated decisions - the architectural pattern for AI that can be trusted when mistakes carry consequences. **Agentic pipeline:** 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Escalation) with conditional routing, tool calling, and dynamic task delegation. **HITL guardrails:** deterministic logic makes every regulated compliance call - LLM extracts and reasons, never decides; confidence-based routing escalates to human engineers when certainty is insufficient. **RAG architecture:** section-aware document ingestion with TF-IDF retrieval and hybrid reranking; citation-backed outputs with 100% citation correctness and 0% unsupported-claim rate on eval set. **Measured outcomes:** 93.6% precision, 87.9% F1, pass@1 81.7% / pass@3 88.3% across 1,240 test cases - evaluation-driven iteration throughout.

AI on FHIR: LLM Integration & Distributed Backend on AWS - Regulated Environment

LLM Integration · AWS Lambda · Python · Distributed · HIPAA · Production

Python · Flask · React.js · TypeScript · Docker · AWS Lambda · PostgreSQL · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Production LLM integration in a regulated environment where AI behavior must be trusted: NLP-powered tool-calling translating natural-language queries (EN/ES) into structured FHIR API calls via LLM; AWS Lambda serverless backend with API gateway; PostgreSQL data services; HIPAA-compliant RBAC with full audit trail on every AI-assisted retrieval; observability stack. Demonstrates LLM integration + distributed backend + compliance on AWS - the engineering combination applicable to high-consequence operational environments. ~60% reduction in manual retrieval time.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017